

Calendar & Clock Problems

COMPLETE LEARNING GUIDE

What You'll Learn

Calendar problems ask for the day of the week on a given date (or how a repeating date pattern shifts across years), using the 'odd days' method built on the fact that the days of the week repeat with a period of 7 and a normal year contains exactly 1 extra ('odd') day beyond 52 complete weeks (2 odd days for a leap year). Clock problems ask about the angle between the hour and minute hands at a given time, or when the hands next align/oppose, using the fixed facts that the minute hand moves 6° per minute and the hour hand moves 0.5° per minute. Both topics are formula-driven rather than puzzle-driven, and speed comes from having the core formulas and the odd-days-per-century reference table memorized cold. This sub-module covers the odd-days method and leap-year rules, finding the day of the week for any date, recurring-date-shift patterns, the clock-angle formula, hands-coincide and hands-opposite problems, and faulty (fast/slow) clock adjustments.

1. The Odd Days Concept and Leap Year Rule

An 'odd day' is the remainder of days beyond complete weeks in a given period. A normal (non-leap) year has $365 \text{ days} = 52 \text{ weeks} + 1 \text{ day}$, so it contributes 1 odd day; a leap year (366 days) contributes 2 odd days. A year is a leap year if divisible by 4, EXCEPT century years (ending in 00), which must be divisible by 400 to be leap years (so 2000 was a leap year, but 1900 was not).

- Is 2024 a leap year? $2024 \div 4 = 506$ exactly, and it's not a century year, so yes, 2024 is a leap year (366 days, 2 odd days). Is 1900 a leap year? It's a century year, and $1900 \div 400$ is not exact, so NO, 1900 is not a leap year despite being divisible by 4.

2. Odd Days Reference Table for Longer Periods

For century-scale calculations, memorize: 100 years = 5 odd days, 200 years = 3 odd days, 300 years = 1 odd day, 400 years = 0 odd days (since 400 years contains exactly 97 leap years, making the total days an exact multiple of 7) — this 400-year-cycle fact is why the Gregorian calendar repeats its day-pattern every 400 years.

- 400 years always contributes 0 odd days: $(400 \times 365 + 97 \text{ leap days}) = 146097 \text{ days}$, and $146097 \div 7 = 20871$ exactly — no remainder, confirming the 400-year cycle repeats the calendar exactly.

3. Finding the Day of the Week for a Given Date

Standard method: compute total odd days from a known reference point (often January 1, year 1, which is defined as a Monday) up to the day before the target date, take that total mod 7, and map the remainder (0=Sunday, 1=Monday, ... 6=Saturday) to get the day of the week — in practice, most problems give a nearby known reference date to calculate FROM, rather than starting from year 1.

- If January 1, 2023 was a Sunday, what day is January 1, 2024? 2023 is not a leap year, contributing 1 odd day. Sunday + 1 odd day = Monday. So January 1, 2024 was a Monday.

4. Recurring Date-Shift Patterns

For 'same date, different year' questions, add the odd days contributed by each intervening year (checking each for leap-year status) and shift the known day of the week by that total mod 7 — this is a direct extension of the day-of-week method, just applied to a multi-year gap.

- If 15th August 2021 was a Sunday, what day is 15th August 2025? Years in between contributing odd days: 2021!2022 (1, not leap), 2022!2023 (1), 2023!2024 (2, 2024 is leap), 2024!2025 (1) = total 5 odd days. Sunday + 5 = Friday. So 15th August 2025 is a Friday.

5. The Clock-Angle Formula

The minute hand moves $360^\circ/60 \text{ min} = 6^\circ$ per minute. The hour hand moves $360^\circ/(12 \times 60) = 0.5^\circ$ per minute. The angle between them at H hours and M minutes is $|30 \times H - 5.5 \times M|$ degrees (taking the result mod 360, and using the smaller of the angle or 360-minus-the-angle if it exceeds 180°) — this single formula covers virtually every 'angle between hands at a given time' question.

- What is the angle between the hands at 3:40? $|30 \times 3 - 5.5 \times 40| = |90 - 220| = 130^\circ$. Since $130^\circ < 180^\circ$, that's already the smaller angle: 130° .

6. Hands Coincide or Are Opposite

The hands coincide (0°) roughly every $65 \frac{5}{11}$ minutes (not exactly every 60, since the hour hand also moves), and are exactly opposite (180°) roughly every $65 \frac{5}{11}$ minutes as well, offset by a half-cycle — use the same angle formula, setting it equal to 0° (coincide) or 180° (opposite) and solving for M given H.

- At what time between 4 and 5 o'clock are the hands exactly opposite (180°)? $|30 \times 4 - 5.5 \times M| = 180$! $120 - 5.5M = \pm 180$. Taking $120 - 5.5M = -180$! $5.5M = 300$! $M = 54.54$ (54 and $6/11$ minutes). So the hands are opposite at approximately 4:54:33.

7. Faulty (Fast/Slow) Clock Adjustments

A clock that gains or loses time runs at a different effective rate than a correct clock — if a clock gains M minutes every H hours of ACTUAL time, its effective rate is $(H \times 60 + M)$ minutes of clock-time per $H \times 60$ minutes of real time; use this ratio to convert between the faulty clock's reading and real elapsed time.

- A clock gains 10 minutes every 24 hours. In real 24 hours (1440 minutes), the faulty clock shows 1450 minutes. If the faulty clock shows exactly 24 hours have passed (1440 clock-minutes), real time elapsed = $1440 \times (1440/1450)$ "H 1430.07 real minutes — very slightly less than 24 real hours, since the clock is running fast and reaches '24:00' before real time does.

Common Pitfalls

- Applying the divisible-by-4 leap year rule to century years without checking the divisible-by-400 exception (e.g., wrongly treating 1900 as a leap year).
- Forgetting that a leap year contributes 2 odd days, not 1, when calculating across a span that includes a leap year.
- Miscounting which years in a date-shift range are leap years, especially when the range crosses a century boundary.
- Forgetting to reduce the hour value to a 12-hour or correct decimal-hour basis before applying the clock-angle formula (e.g., using 15 instead of 3 for 3 PM without adjusting).
- Taking the raw output of the angle formula without checking whether it exceeds 180° and needs to be subtracted from 360° to get the actual (smaller) angle asked for.
- Assuming clock hands coincide exactly every 60 minutes instead of the correct $\sim 65 \frac{5}{11}$ minutes, due to the hour hand's own continuous movement.
- In faulty-clock problems, applying the gain/loss ratio in the wrong direction (converting real time to clock time when the question asks the reverse, or vice versa).

Key Takeaways

- A century year (ending in 00) is a leap year only if divisible by 400, overriding the normal divisible-by-4 rule.
- Odd days per year: 1 for a normal year, 2 for a leap year — always check leap-year status for every year spanned.
- 400 years always contribute exactly 0 odd days — the basis of the Gregorian calendar's 400-year repeat cycle.
- Clock-angle formula: $|30H - 5.5M|$ degrees, adjusted to the smaller angle ($\text{"d}180^\circ$) if the raw result exceeds 180° .
- Clock hands coincide or oppose roughly every $65 \frac{5}{11}$ minutes, not every 60 — because the hour hand keeps moving too.
- For faulty clocks, convert between real time and clock-shown time using the ratio of actual elapsed time to clock-time elapsed, based on the stated gain/loss rate.

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IndiaBix — Calendar

<https://www.indiabix.com/aptitude/calendar/>

GeeksforGeeks — Clock Aptitude Questions

<https://www.geeksforgeeks.org/clock-based-questions-aptitude/>